00 Lab
PPS Lab Introduction and Setup

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C.D.L. Magistrale in Ingegneria e Scienze Informatiche
Alma Mater Studiorum—Università di Bologna, Cesena

a.a. 2025/2026

On how to follow lab lessons

- Students receive lab materials (slides and code) in advance (ideally Friday, latest Sunday)
- Teachers and tutors are available for the entire duration of the lab
  - exercise assignation: students will try to solve them
- Students can require dedicated interactions with the teacher and the tutor
  - to discuss and receive feedback on proposed solutions, ...
- This first lesson is dedicated to the setup of the development environment,
  - Configure IntelliJ IDEA (most important); make sure the JDK and IDE are properly set up
    - You can install the Scala plugin now, but it will be mainly useful for the next lessons

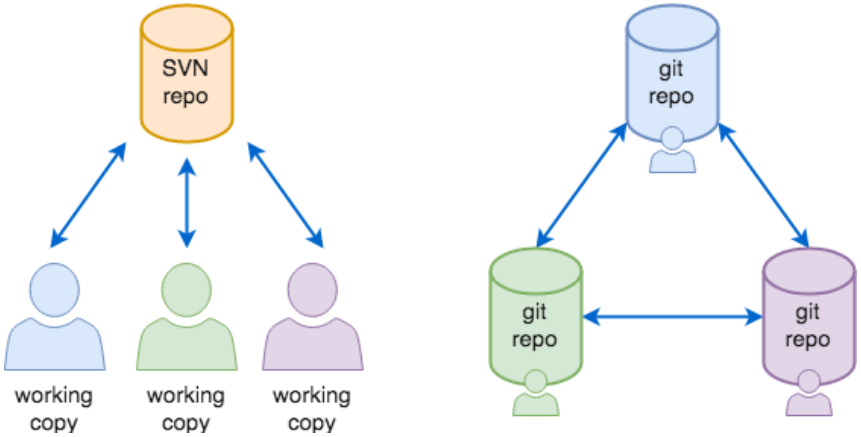
Outline

- A brief recall on DVCS and Git
- Setting up IntelliJ Idea (and Scala Plugin)

git: brief intro (i)

Basics on version control

- **Version Control System (VCS)**: system for the management of "versions" of data
- **Repository**: stores a set of files and the history of their versions (or the history changes made to them)
- **Centralized VCSs**: single remote repo + working copies
  - A **working copy** is a local copy of the repo
- **Distributed VCSs**: each developer has a copy of the entire repository



git: brief intro (ii)

Basics on git

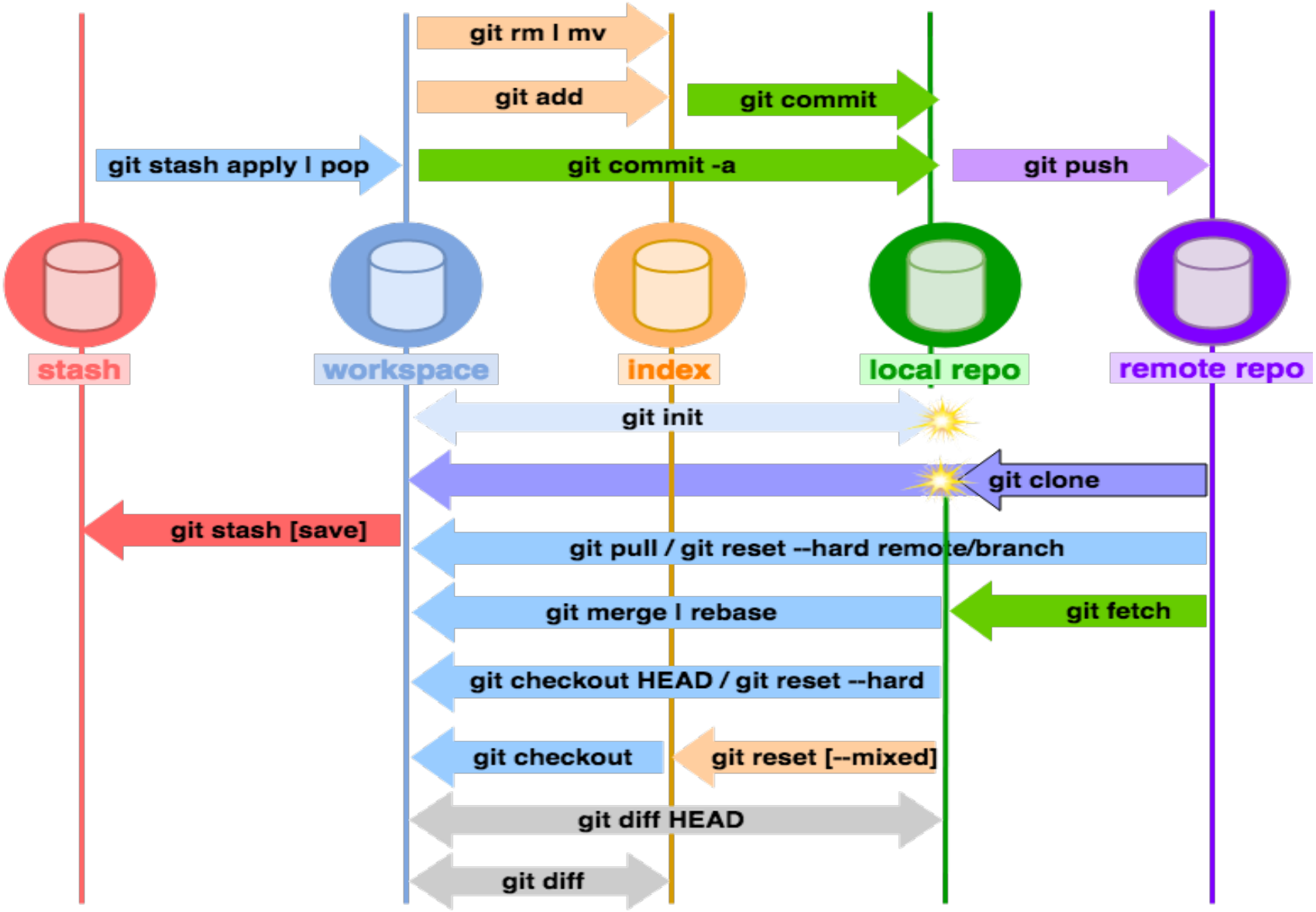
- git thinks of data as a stream of **snapshots** (rather than a list of file-based changes)
  - But does use **deltas** for storage!
- **Git directory**: stores metadata and object DB (commits, branches..)
- **Working tree**: a filesystem tree containing a single **checkout** of one version of the project (+ local changes)
- **Staging area** (aka **index**): keeps track of the snapshot (actually, delta w.r.t. HEAD) that will go into the next commit
- Files in working tree can be in two states: **tracked** or **untracked**
- Tracked files can be: **unmodified** or **modified** and/or **staged**
- **Branch**: independent line of development; new commits are recorded in the history of current branch
- **HEAD**: pointer to the current snapshot (branch or commit)

git: brief intro (iii)

Key commands

- `init`: initialize a new git repository
- `clone`: downloads a project and its entire version history
- `status`: shows untracked files, modified files, staged files
- `add`: propose changes by adding them as a snapshot to the **index** (the staging area); also resolves merge-conflicted files
- `diff`: shows difference between a commit and the working tree, or between the working tree and the index
- `commit` **(check in)**: the changes are committed to the HEAD
- `push`: send local changes to another (possibly remote) repository
- `checkout`: updates the working tree by switching to a branch/commit
- `branch`: manages branches (list, create, delete, . . .)
- `fetch`: download objects and references from another repository
- `pull`: incorporates changes from a remote repository into the current branch
- `push`: update remote references along with associated objects
- `merge`: incorporates changes from a branch into the current branch

git: a pictorial view



git: workflows (i)

Simple workflow

- Create and/or clone a remote repository
- Create or select a branch of choice
- Create and/or modify files in the working tree
- Stage your changes
- Commit your changes
- When you are done, merge your branch into the main branch (master)
  - If any conflict, resolve them, add (mark) them as resolved and commit
- Push your changes to the remote
  - May fail if someone pushed new stuff; in this case, pull and incorporate changes before retrying the push.

git: workflows (ii)

Workflow for practice in lab (i)

Introduce yourself to git

- `$ git config --global user.name "Name"`
- `$ git config --global user.email your@email.com`

Lab workflow

- We give you a remote repository with code for lab
  - e.g., `https://github.com/unibo-pps/pps-lab00`
- If it is the first time, you clone the repo
  - `$ git clone git@github.com:unibo-pps/pps-24-25-lab00.git`

  otherwise you pull from the remote
  - `$ git pull origin` (origin: default remote name for a cloned repo)
- Create or modify files in the working tree, stage your changes, and commit them
  - `$ git add .`
  - `$ git commit -m "Commit msg"`

On forks (i)

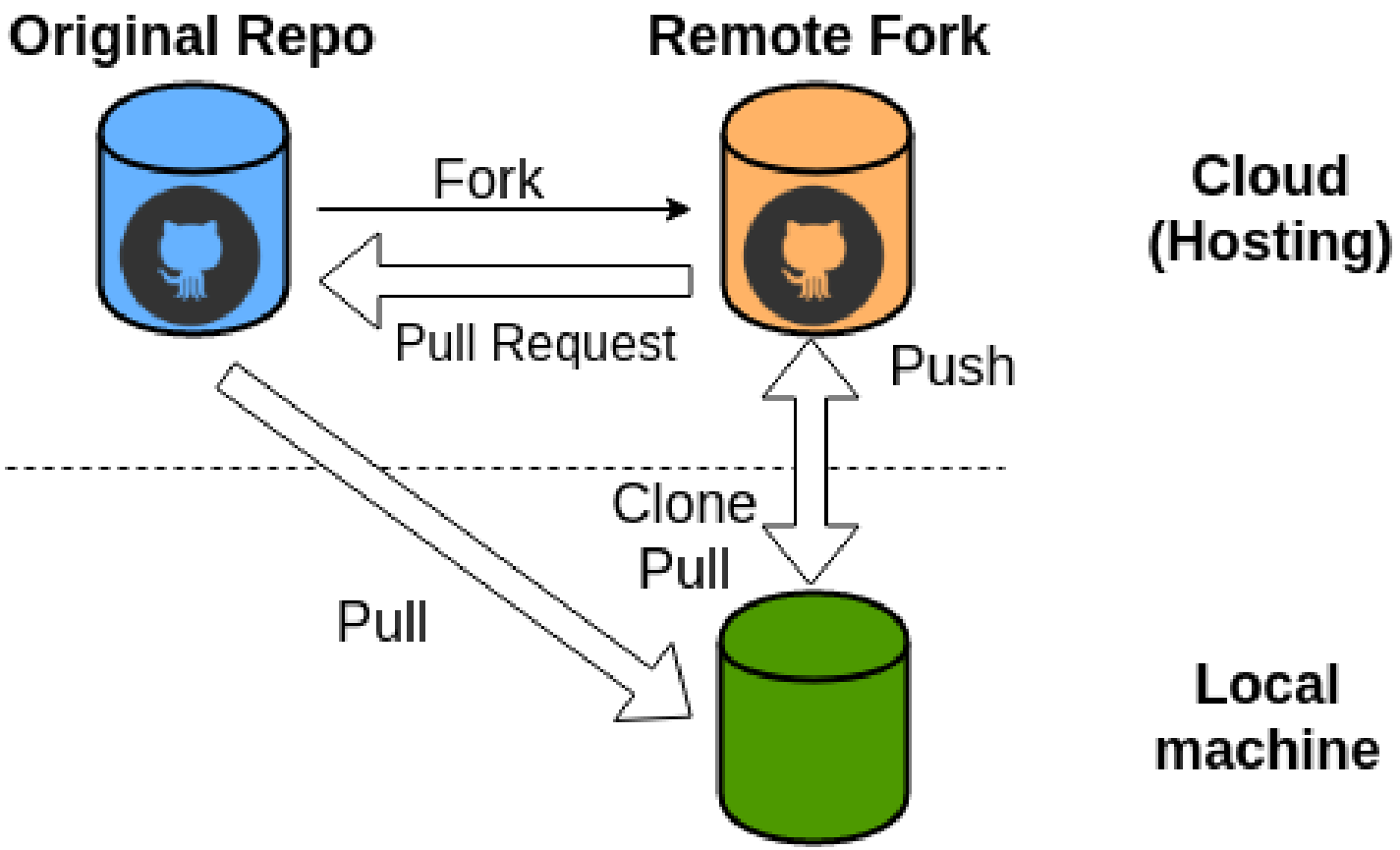
You may want to keep your own remote repo and push changes to it

Forking

a) Create a new, empty remote repo
b) Explicitly create a fork (e.g., on GitHub or Bitbucket)
   - Preferred, as this supports *pull requests*

- Clone locally your own remote fork
- Add the remote referring to the original repo:
  `$ git remote add sync <originalRepo>`
- Sync with the original repo `$ git pull sync master`
- Set up a local branch to track the original one (for pulls)
  `$ git branch originalMaster --track sync/master`
- Push into the forked repo and track it `$ git push -u origin master`
- Use `sync` to fetch from the original repo
- Use `origin` to pull and push to your forked repo

On forks (ii)


Original Repo
Remote Fork
Fork
Pull Request
Push
Cloud
(Hosting)
Clone
Pull
Pull
Local
machine

Setting up the development environment

To install

- git
  - https://git-scm.com/downloads
  - more on how to use git: https://git-scm.com/book/en/v2/Getting-Started-First-Time-Git-Setup
- JDK 21 (suggested)
  - installation using adoptium https://adoptium.net/
- IntelliJ IDEA Community/Professional Edition 2025.2
  - https://www.jetbrains.com/idea/download/?section=linux
  - As Unibo students you can get Professional for free with your university email at https://www.jetbrains.com/shop/eform/students
- Scala Plugin for IntelliJ IDEA
  - https://www.jetbrains.com/help/idea/discover-intellij-idea-for-scala.html
- Scala 3.3 (latest stable version)
  - https://www.scala-lang.org/download/scala3.html

First steps in IntelliJ IDEA (i)

Key notions in IntelliJ

- Project settings: `https://www.jetbrains.com/help/idea/`
- Core concepts:
  - **Project**: Complete software solution with settings in `.idea` folder
  - **Module**: Independent unit within a project that can be built/run separately
  - **SDK**: Required development kit (e.g., Java SDK) configured via `File | Project Structure`
- Version control: **Do not** include any IDE-specific files in the repo
  - **.idea** folder: IDE settings
  - **.iml** files: module settings

Lab Setup

- Clone (or fork and clone) the repository at `https://github.com/unibo-pps/pps-lab00`
- Open the project in IntelliJ Idea
  - File — Open — select the folder of the cloned project
- It should correctly import the project and set up the SDK
  - We mainly leverage external building tools (sbt, maven, gradle) for building and running the project, reducing the need for IDE-specific settings

Setup "Exercise"

1. Analyse the proposed code to understand the application logic
2. Run the application by clicking Run near the main method in `src/main/java/Main.java`
3. Try to modify/refactor some parts of the proposed code to become familiar with the IDE
4. Run the proposed test suite (see `test/model/PersonTest.java`) by the green arrow near the class name
   - Run all tests using the green arrow near in the class name
5. Try to add other simple tests for the application